

Submission PHYS1-FINAL-SUB01

Question PHYS1-FINAL-Q01

scratch: show step

1. $v = v_i + at = 8 + 2(5) = 18 \text{ m/s}$

2. $\Delta x = v_i t + \frac{1}{2}at^2 = 8(5) + \frac{1}{2}(2)(5)^2 = 65 \text{ m}$

=> FINAL: $v = 18 \text{ m/s}$; displacement = 65 m (check units/interpretation)

Question PHYS1-FINAL-Q02

scratch: show step

1. $N = mg = 19.6 \text{ N}$; friction = $\mu N = 3.9 \text{ N}$

2. $\Sigma F = 13 - 3.9 = 9.1 \text{ N}$

3. $a = \Sigma F / m = 4.54 \text{ m/s}^2$

=> FINAL: $a = 4.54 \text{ m/s}^2$; friction opposes motion (check units/interpretation)

Question PHYS1-FINAL-Q03

scratch: show step

1. $mgh = \frac{1}{2}mv^2 \Rightarrow v = \sqrt{2gh} = \sqrt{2 \cdot 9.8 \cdot 6}$

2. $v = 10.84 \text{ m/s}$; mass cancels

=> FINAL: $v = 10.84 \text{ m/s}$ (check units/interpretation)

Question PHYS1-FINAL-Q04

scratch: show step

1. $p_{\text{total, before}} = p_{\text{total, after}}$

2. It applies when external impulse on the system is negligible.

=> FINAL: total momentum before equals after when external impulse is negligible (check units/interpretation)

Question PHYS1-FINAL-Q05

scratch: show step

1. Impulse is area under $F(t)$: $J = F\Delta t = 6(0.5) = 3 \text{ N}\cdot\text{s}$.

2. $\Delta p = J$, so momentum changes by $3 \text{ kg}\cdot\text{m/s}$ in the force direction.

=> FINAL: impulse = $3 \text{ N}\cdot\text{s}$; momentum changes by $3 \text{ kg}\cdot\text{m/s}$ (check units/interpretation)

Question PHYS1-FINAL-Q06

scratch: show step

1. $v = v_i + at = 5 + 2(5) = 15 \text{ m/s}$

2. $\Delta x = v_i t + \frac{1}{2}at^2 = 5(5) + \frac{1}{2}(2)(5)^2 = 50 \text{ m}$

=> FINAL: $v = 15 \text{ m/s}$; displacement = 50 m (check units/interpretation)

Question PHYS1-FINAL-Q07

scratch: show step

1. $N = mg = 68.6 \text{ N}$; friction = $\mu N = 13.7 \text{ N}$

2. $\Sigma F = 23 - 13.7 = 9.3 \text{ N}$

3. $a = \Sigma F / m = 1.33 \text{ m/s}^2$

=> FINAL: $a=1.33 \text{ m/s}^2$; friction opposes motion (check units/interpretation)

Question PHYS1-FINAL-Q08

scratch: show step

1. $mgh = \frac{1}{2}mv^2 \Rightarrow v = \sqrt{2gh} = \sqrt{2 \cdot 9.8 \cdot 5}$

2. $v=9.90 \text{ m/s}$; mass cancels

=> FINAL: $v=9.90 \text{ m/s}$ (check units/interpretation)

Question PHYS1-FINAL-Q09

scratch: show step

1. $p_{\text{total,before}} = p_{\text{total,after}}$

2. It applies when external impulse on the system is negligible.

=> FINAL: total momentum before equals after when external impulse is negligible (check units/interpretation)

Question PHYS1-FINAL-Q10

scratch: show step

1. Impulse is area under $F(t)$: $J = F\Delta t = 6(0.5) = 3 \text{ N}\cdot\text{s}$.

2. $\Delta p = J$, so momentum changes by $3 \text{ kg}\cdot\text{m/s}$ in the force direction.

=> FINAL: impulse = $3 \text{ N}\cdot\text{s}$; momentum changes by $3 \text{ kg}\cdot\text{m/s}$ (check units/interpretation)